UNIVERSITY INSTITUTE OF COMPUTING

PROJECT REPORT
ON
ONLINE FOOD DELIVERY TIME & RATING ANALYSIS

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Submitted by:

Name: Jyoti Kumari

UID: 23BCA10071

Section: BCA – 5 'A'

Submitted to:

Name: Harkamal Kaur

Designation: Assistant-Professor

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1. Introduction

The rapid growth of the internet and smartphones has transformed the way people order food. Online food delivery platforms such as Zomato, Swiggy, and others have become an integral part of urban life across India. Millions of orders are placed daily, generating enormous volumes of data including delivery times, customer ratings, order amounts, and geographic information.

This mini project, undertaken as part of the Data Interpretation course, focuses on analysing a structured dataset of 20 online food delivery orders. Using Microsoft Excel as the primary analytical tool, we apply ten different built-in functions to extract meaningful patterns and business insights from the raw data.

The study simulates a real-world scenario faced by food delivery platform managers and restaurant partners who need quick, data-driven answers to operational questions — such as which cities have the highest revenue, which food categories attract the best ratings, and whether delivery speed correlates with customer satisfaction.

The dataset spans four cuisine types — Indian, Chinese, Italian, and Fast Food — and includes key metrics such as actual and estimated delivery times, customer ratings, order values, and distance. The project concludes with a Summary Dashboard that presents consolidated key performance indicators (KPIs) for quick managerial review.

2. Aim

To design and analyses a comprehensive Excel-based case study on Online Food Delivery data by building structured tables, applying advanced Excel formulas, and deriving actionable insights related to delivery performance and customer satisfaction.

3. Objectives

The specific objectives of this mini project are:

1. To create a well-structured Excel workbook with realistic online food delivery data (20 rows).
2. To demonstrate the use of 10 different Excel functions — SUM, AVERAGE, MAX, MIN, COUNT, COUNTIF, AVERAGEIF, IF, RANK, and SUMIF — on the dataset.
3. To calculate key performance indicators (KPIs) such as total revenue, average delivery time, and average rating.
4. To perform category-wise and city-wise analysis using conditional formulas.
5. To build a clean KPI Dashboard sheet within the workbook for visual summarization.
6. To interpret the results and draw meaningful conclusions about delivery efficiency and customer satisfaction.
7. To strengthen practical data interpretation skills using a real-world business scenario.

4. Scope of the Study

The scope of this project is defined by the following boundaries and inclusions:

4.1 Inclusions

- Dataset of 20 food delivery orders across 8 major Indian cities (Delhi, Mumbai, Bangalore, Chennai, Hyderabad, Pune, Jaipur, Kolkata).
- 12 data attributes per order: Order ID, Customer Name, Restaurant Name, Food Category, Order Amount (₹), Delivery Time (minutes), Distance (km), Rating (1–5), Delivery Partner, City, Order Date, and Order Status.
- Application of 10 Excel functions spanning statistical, logical, and lookup categories.
- City-wise aggregation using COUNTIF, SUMIF, and AVERAGEIF formulas.
- A dedicated KPI Dashboard sheet for executive-level summarization.

4.2 Limitations

- The dataset is illustrative and limited to 20 records; real-world platforms handle millions of transactions.
- No machine learning or predictive modelling is applied — analysis is purely descriptive.
- Weather, traffic, and seasonal factors are not captured in the dataset.
- The study uses Microsoft Excel only; no database or BI tools (Power BI, Tableau) are in scope.

5. Problem Definition

Online food delivery platforms generate vast amounts of transactional data daily. However, raw data in itself is of little value unless it is organized, cleaned, and analyzed systematically. Restaurant managers, platform operations teams, and marketing analysts face the following recurring challenges:

5.1 Key Business Problems

1. Lack of visibility into delivery performance: Managers cannot easily identify which delivery partners or cities are underperforming without aggregated data.
2. Inability to measure customer satisfaction by category: Average ratings are known overall but not broken down by food type (e.g., Biryani vs. Fast Food).
3. No quick way to classify orders: There is no automated system to flag an order as 'Fast Delivery' or 'Moderate Delivery' based on time taken.
4. Revenue leakage by category: The revenue contribution of individual food categories is unknown without conditional calculations.
5. Difficulty in prioritizing cities for expansion: Without city-wise aggregation (orders, revenue, ratings), strategic decisions are guesswork.

5.2 Problem Statement

"Given a dataset of online food delivery orders, how can we use Excel functions to compute delivery KPIs, classify orders, aggregate revenue by category and city, and rank orders — thereby enabling data-driven operational decisions?"

6. Existing Gaps in Practice

Despite the wide availability of food delivery data, several gaps exist in how small restaurants and platform operators currently manage and interpret this data:

Sr.	Gap Area	Current Situation	Impact
1	**No Centralized Data Repository**	Orders stored in POS systems or paper logs	Difficult to run analysis; data silos
2	**Manual Rating Aggregation**	Ratings noted individually, no automated avg	Time-consuming; prone to error
3	**No Category-wise Revenue Tracking**	Total revenue tracked but not by food type	Cannot identify top-revenue categories
4	**Absence of Delivery Speed Classification**	Delivery time noted but not benchmarked	No performance standards for partners
5	**City-level Performance Blind Spots**	National totals available, city data absent	Misallocation of marketing budgets
6	**No Order Ranking System**	No way to identify top or bottom orders	High-value customers not rewarded
7	**Reactive Decision Making**	Decisions made after problems escalate	Late interventions increase costs

This project addresses gaps 2 through 6 directly using Excel formulas, providing a simple, accessible, and replicable solution that any small business can adopt without expensive software.

7. Dataset Description

The dataset used in this project is a simulated but realistic representation of food delivery orders placed across major Indian cities. It is structured as a flat table in Microsoft Excel with the following specifications:

- Total Records: 20 orders
- Number of Columns: 12 attributes

- Date Range: January 5–24, 2024
- Cities Covered: Delhi, Mumbai, Bangalore, Chennai, Hyderabad, Pune, Jaipur, Kolkata
- Food Categories: Biryani, Fast Food, Pizza, BBQ, South Indian, Wraps, Thali, Chinese, Continental, Snacks, Sandwiches, Momos, Punjabi

7.1 Sample Data (First 8 Records)

Order ID	Customer	Restaurant	Category	Amt (₹)	Time (min)	Dist (km)	Rating	Status
ORD001	Rahul Sharma	Zomato Kitchen	Biryani	450	35	3.2	4.5	**Delivered**
ORD002	Priya Singh	Burger King	Fast Food	320	28	2.1	4.0	**Delivered**
ORD005	Vikas Gupta	KFC	Fast Food	380	25	1.9	4.6	**Delivered**
ORD007	Rohit Mehta	Barbeque Nation	BBQ	950	55	6.5	3.5	**Cancelled**
ORD015	Kiran Malhotra	Ammi's Biryani	Biryani	410	32	3.0	4.9	**Delivered**
ORD017	Ritu Kapoor	Pind Balluchi	Punjabi	830	60	7.2	3.2	**Cancelled**
ORD019	Ankita Mishra	Wow! Momo	Momos	260	19	1.1	4.6	**Delivered**
ORD020	Vikram S.	Behrouz Biryani	Biryani	720	48	6.0	4.8	**Delivered**

8. Excel Formulas — Code, Explanation & Output

All formulas below are entered in the 'Analysis & Formulas' sheet of the workbook. They reference the raw data in the 'Order Data' sheet (cells E3:E22, F3:F22, etc.). The results shown are calculated outputs after applying each formula to the 20-row dataset.

Sr.	Formula	Excel Code	Output	Interpretation
1	SUM	=SUM('Order Data'!E3:E22)	₹9,730	Total revenue collected from all 20 orders.
2	AVERAGE	=AVERAGE('Order Data'!F3:F22)	33.9 min	On average, each delivery takes ~34 minutes.
3	MAX	=MAX('Order Data'!E3:E22)	₹950	Highest single order was ₹950 (Barbeque Nation).
4	MIN	=MIN('Order Data'!F3:F22)	18 min	Fastest delivery completed in 18 minutes.
5	COUNT	=COUNT('Order Data'!E3:E22)	20	Confirms 20 valid order records exist.
6	COUNTIF	=COUNTIF('Order Data'!L3:L22,"Delivered")	17	17 out of 20 orders were successfully delivered.
7	AVERAGEIF	=AVERAGEIF('Order Data'!D3:D22,"Biryani','Order Data'!H3:H22)	4.73/5	Biryani orders receive the highest average rating.
8	IF	=IF(AVERAGE(...)<=30,"Fast Delivery","Moderate Delivery")	Moderate	Avg time 33.9 min > 30 min → tagged Moderate.
9	RANK	=RANK(E3,'Order Data'!E3:E22,0)	16	ORD001 (₹450) ranks 16th among all orders by amount.
10	SUMIF	=SUMIF('Order Data'!D3:D22,"Fast Food",'Order Data'!E3:E22)	₹990	Fast Food category generated ₹990 total revenue.

8.1 Detailed Formula Walkthroughs

Formula 1: SUM — Total Revenue

```
=SUM('Order Data'!E3:E22)
```

The SUM function adds all values in the Order Amount column (E3 to E22). This gives the platform's total gross revenue across all 20 orders. In a real scenario, this formula would scale to thousands of rows without modification.

Formula 2: AVERAGE — Mean Delivery Time

```
=AVERAGE('Order Data'!F3:F22)
```

AVERAGE computes the arithmetic mean of all delivery times. The result (33.9 min) tells management that most orders take just over half an hour. This can be used to set SLA (Service Level Agreement) benchmarks for delivery partners.

Formula 3 & 4: MAX and MIN

```
=MAX('Order Data'!E3:E22)
```

```
=MIN('Order Data'!F3:F22)
```

MAX identifies the single highest-value order, useful for VIP customer identification. MIN pinpoints the fastest delivery — Wow! Momo at 18 minutes and just 1.1 km distance, demonstrating how proximity directly impacts speed.

Formula 5 & 6: COUNT and COUNTIF

```
=COUNT('Order Data'!E3:E22)
```

```
=COUNTIF('Order Data'!L3:L22,"Delivered")
```

COUNT verifies the dataset size. COUNTIF filters by condition — finding that 17 of 20 orders (85%) were delivered successfully, 2 were cancelled, and 1 was delayed. This is a core metric for platform reliability reporting.

Formula 7: AVERAGEIF — Conditional Average

```
=AVERAGEIF('Order Data'!D3:D22,"Biryani",'Order Data'!H3:H22)
```

AVERAGEIF calculates the average rating, but only for rows where the Food Category equals 'Biryani'. The result (4.73/5) confirms that Biryani orders are the highest-rated food category — valuable intelligence for menu promotion decisions.

Formula 8: IF — Logical Classification

```
=IF(AVERAGE('Order Data'!F3:F22)<=30,"Fast Delivery","Moderate Delivery")
```

The IF function applies a business rule: if average delivery time is 30 minutes or less, classify as 'Fast'; otherwise 'Moderate'. This automated classification can drive dynamic displays on dashboards, replacing manual decision-making.

Formula 9: RANK — Order Ranking

```
=RANK('Order Data'!E3,'Order Data'!E3:E22,0)
```

RANK allows each order to be assigned a position relative to all others. Applied to every row, it enables identification of the top 5 highest-value orders for loyalty rewards. The descending sort (0) ensures the highest amount gets rank 1.

Formula 10: SUMIF — Category Revenue

```
=SUMIF('Order Data'!D3:D22,"Fast Food",'Order Data'!E3:E22)
```

SUMIF sums only the order amounts where food category is 'Fast Food'. The ₹990 result can be compared against Biryani (₹1,580) or Pizza (₹1,220) to determine

which category is the highest revenue contributor — critical for menu investment decisions.

8.2 City-Wise Analysis Output

The following results were produced by applying COUNTIF, SUMIF, and AVERAGEIF formulas with city as the condition:

City	Orders	Revenue (₹)	Avg Rating
Delhi	4	1,990	4.3
Mumbai	3	2,320	3.9
Bangalore	2	1,400	4.3
Chennai	2	840	4.2
Hyderabad	1	290	4.7
Pune	1	540	4.2
Jaipur	1	210	4.8
Kolkata	1	750	3.7

9. KPI Dashboard — Final Outputs

The KPI Dashboard sheet consolidates the most important metrics into a visual card layout. Each KPI card uses a live Excel formula so values update automatically when source data changes.

KPI Metric	Formula Used	Result
Total Orders	=COUNT(E3:E22)	**20**
Total Revenue	=SUM(E3:E22)	**₹9,730**
Average Delivery Time	=AVERAGE(F3:F22)	**33.9 minutes**
Average Customer Rating	=AVERAGE(H3:H22)	**4.23 / 5.0**
Fastest Delivery	=MIN(F3:F22)	**18 minutes**
Highest Order	=MAX(E3:E22)	**₹950**
Delivered Orders	=COUNTIF(L3:L22,"Delivered")	**17 (85%)**

Cancelled Orders	=COUNTIF(L3:L22,"Cancelled")	**2 (10%)**

10. Interpretation of Results

- Delivery Efficiency: With an average delivery time of 33.9 minutes, the platform is in the 'Moderate' speed band. Implementing route optimization for distances above 5 km could bring this below the 30-minute target.
- Customer Satisfaction: An average rating of 4.23/5 indicates strong overall satisfaction. Biryani (4.73) and Jaipur orders (4.8) are standout performers — signaling that specialty cuisines and shorter distances improve experience.
- Revenue Concentration: Delhi and Mumbai together account for over 45% of total revenue, suggesting these are key markets that need consistent service quality.
- Cancellation Rate: 10% cancellation (2 orders) warrants investigation. Both cancelled orders had delivery times above 55 minutes, suggesting time delays trigger cancellations.
- Category Revenue: Biryani (₹1,580) and BBQ (₹950 single order) lead revenue per order, making them high-priority categories for promotional investment.
- Rank Analysis: RANK formula reveals that the top 5 orders by value range from ₹720 to ₹950, all from premium restaurant categories (Barbeque Nation, Pind Baluchi, Behrouz Biryani).

11. Conclusion

This mini project successfully demonstrates the power of Microsoft Excel as a data interpretation tool for real-world business scenarios. By applying 10 different Excel functions — SUM, AVERAGE, MAX, MIN, COUNT, COUNTIF, AVERAGEIF, IF, RANK, and SUMIF — to a dataset of 20 online food delivery orders, we were able to:

1. Compute total and category-wise revenue with SUM and SUMIF.
2. Measure average delivery performance and rating with AVERAGE and AVERAGEIF.
3. Identify extreme values (best and worst orders) using MAX and MIN.
4. Count and classify orders by delivery status using COUNT and COUNTIF.

5. Automate delivery speed classification using logical IF functions.
6. Rank orders to identify high-value transactions using RANK.

The analysis revealed that the platform achieves an 85% delivery success rate, an average rating of 4.23/5, and generates the highest revenue from Biryani and Fast-Food categories. Delhi and Mumbai are the top revenue cities. Cancellations appear correlated with long delivery times, pointing to a need for better logistics in high-distance orders.

This project validates that even without expensive analytics software, Excel-based data interpretation can generate actionable business intelligence. The skills practiced here — formula construction, conditional aggregation, logical classification, and dashboard design — are foundational for any data analyst or business manager working in e-commerce, logistics, or food-tech industries.

Key Takeaway: Structured data, combined with the right Excel formulas, transforms a simple spreadsheet into a powerful decision-support tool — enabling faster, evidence-based operational decisions in the competitive food delivery industry.

12. References

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